Jaskirat Singh

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Personal Profile

Physics student working at the intersection of nuclear structure, particle interactions, and nonlinear dynamical systems. I have a strong interest in using computational and data-driven methods for complex physical systems. I am developing physics-informed neural frameworks to model structured residual behaviour and analyse spectral signatures in nonlinear systems. My background includes scattering theory, PIXE and muon interactions, as well as particle-matter interaction processes.

Education

2023 - Present: BS–MS Integrated Program in Physics (Major) with Minor in Data Science

IISER Mohali, India
Merit-Cum-Means scholarship
CGPA: 8.91/10.00
Relevant Coursework: Quantum Mechanics, Quantum Field Theory I, Nuclear Physics, Relativity, Mathematical Methods for Physicists, Computational Physics, Plasma Physics.

2022 - 2023: Senior Secondary (Class XII)

S.R.M. International School, Hanumangarh, India
Percentage(PCM): 97.66/100.00
Honors: Physics and Mathematics Subject Topper.

Academic Publication

Hierarchical Neural Filtering of Nuclear Mass Residuals and Spectral Signatures of Quantum Chaos

Jaskirat Singh, *Chong Qi*
arXiv:2606.08546 [nucl-th]. Submitted to *Physical Review C*; under peer review.

Research Experience & Teaching

May 2025 – May2026: Hierarchical Neural Filtering of Nuclear Mass Residuals and Spectral Signatures of Quantum Chaos

Under the supervision of Dr. Chong Qi, KTH Royal Institute of Technology, Sweden

- Developed a physics-informed neural network framework to analyze residual structure in nuclear mass deviations.
- Designed and implemented an original hybrid physics-informed neural framework to model nuclear mass systematics and analyze structured residual behavior.
- Investigated frequency components and potential quantum-chaotic signatures in post-correction residuals.

October 2024 - July 2025: Scattering of Subatomic Particles and Related Phenomena

Under the supervision of Dr. Satyajit Jena, IISER Mohali, India

- Worked on classical and quantum-mechanical formulations of scattering phenomena, deriving Rutherford cross-sections and modeling alpha decay via the WKB framework for Coulomb barrier penetration.
- Analyzed Particle Induced X-ray Emission (PIXE) using PWBA-based ionization cross-sections, and performed a comparative study of proton- and muon-induced interactions, examining differences in ionization probability and energy deposition through the Bethe–Bloch formalism.
- Investigated muon-induced processes as an alternative probe to conventional muon tomography, evaluating their feasibility through cross-section and stopping power analysis.

Jan 2026 - May2026: PHY204(Modern Physics) Tutor
Help Sessions for the BS-MS 2024 Batch in Modern Physics, IISER Mohali, India

- Taught concepts of Relativity, Quantum Mechanics, Nuclear & Particle Physics.

Academic Presentations

2026: Entanglement-Assisted Classical Capacity of Noisy Quantum Channels
Graduate Literature Presentation (PHY631), IISER Mohali

- Presented the landmark work of Bennett, Shor, Smolin and Thapliyal (Physical Review Letters, 1999) on the entanglement-assisted classical capacity theorem.
- Covered Shannon information, mutual information, quantum channels, depolarizing and erasure channels, superdense coding, quantum teleportation, and exact capacity formulas.

2026: Ancilla-Assisted Protection of Information: Application to Atom–Cavity Systems
Scientific Poster, Technical Report and Oral Presentation (IDC352), IISER Mohali

- Presented the ancilla-assisted quantum information protection protocol proposed by Gangwar *et al.* (2023).
- Discussed decoherence-free subspaces, cavity quantum electrodynamics, STIRAP-based quantum control, and deterministic recovery of quantum information from environmental noise.

Skills

Programming: Python (scientific computing & ML).

Machine Learning: PyTorch, TensorFlow/Keras, PyTorch Geometric, scikit-learn.

Scientific Computing: NumPy, Pandas, Scipy, Matplotlib.

Research Tools: Git, GitHub, GitHub Pages, Jupyter Notebook, VS Code, LaTeX.

Embedded Systems & Instrumentation: Arduino-based motor control and digital logic interfacing (555 timer, SR latch, logic ICs), L298N driver integration, stepper motor control, oscilloscope signal characterization.

Languages: English (Fluent), Hindi(Native), Punjabi(Native), Rajasthani(Native).

Research Interests

Nuclear properties and systematic patterns in nuclear masses; nonlinear dynamics and complex systems; dynamics of magnetically confined fusion systems; physics-informed machine learning for structured physical modeling; particle scattering and particle-matter interactions.

References

Dr. Chong Qi
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Dr. Satyajit Jena
IISER Mohali, India
Email: sjena@iisermohali.ac.in